

P01. A Priority-Aware Replanning and Resequencing Framework for Coordination of Connected and Automated Vehicles

中英双语博士精读笔记

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2021 · arXiv preprint

Theme / 主题: Scheduling/resequencing + optimal control

Method family / 方法族: Priority-aware resequencing + decentralized optimal control / 优先级重排序与分散最优控制

PDF pages / 页数: 6 **Relevance / 相关度:** 94/100

Source / 来源: <https://arxiv.org/abs/2109.05573>

1 论文全景速览与核心贡献 / High-Level Overview & Core Contributions

1.1 研究背景与痛点 / Background & Pain Points

这篇论文位于“Scheduling/resequencing + optimal control”方向。对你的 JSSP-based Intersection Management 课题，它回答的是高层调度、低层轨迹平滑、安全过滤或真实验证中的一个关键问题：如何让车辆在路口少停、少冲突、少能耗，同时保持在线可执行。

The paper belongs to the theme of Scheduling/resequencing + optimal control. For a JSSP-based intersection-management thesis, it illuminates one of the key layers: scheduling, smoothing, safety filtering, or real-world validation.

Pain point / 痛点: Assumes connected automated vehicles, known paths, and simplified dynamics; the paper is closer to resequencing theory than to learned graph scheduling.

1.2 核心科学问题 / Core Scientific Question

当车辆到达扰动使原始通行序列不再高效或可行时，如何在不牺牲安全约束的前提下快速重排序并生成可执行轨迹？

How can an intersection manager resequence disturbed arrivals while preserving safety and low-level trajectory feasibility?

1.3 科学贡献 / Scientific Contributions

- 方法贡献 (method contribution): Event-driven replanning and priority-aware resequencing layered on decentralized CAV optimal control at a signal-free intersection.

- 安全/避碰贡献 (safety contribution): Uses rear-end and lateral safety constraints around conflict points, with replanning when disturbances change feasibility.
- 平滑/停止避免贡献 (smoothing contribution): The lower-level optimal-control trajectory is constructed so vehicles can conserve momentum rather than stop at the intersection when a feasible exit time exists.
- 创新力度评判 (reviewer judgement): 强相关但中等创新: 把 resequencing 明确嵌入 CAV 最优控制闭环, 贡献主要在系统架构和事件触发逻辑。

1.4 核心概念对照 / Core Concepts

中文概念	English Concept	Role / 学术作用
自动交叉口管理	Autonomous Intersection Management	把信号控制替换为车辆级调度与控制。
轨迹平滑	Trajectory Smoothing	减少急加减速、停车和能耗峰值。
安全约束	Safety Constraints	把碰撞风险写成距离、时间间隔或屏障函数。
Priority-aware resequencing + decentralized optimal control	优先级重排序与分散最优控制	该论文的核心方法族。

2 理论方法论深度解构 / Methodology & Theoretical Framework

2.1 问题数学建模 / Mathematical Formulation

双积分车辆模型与能量型最优控制 / Double-integrator vehicle model with energy-like optimal control

$$\min_{u_i(\cdot), t_i^f} J_i = \frac{1}{2} \int_{t_i^0}^{t_i^f} u_i(t)^2 dt, \quad \dot{p}_i(t) = v_i(t), \quad \dot{v}_i(t) = u_i(t)$$

Symbol / 符号	含义	Meaning	Role / 建模作用
i	车辆索引	vehicle index	把多车协调拆成单车最优控制子问题
p_i(t)	沿路径位置	path-coordinate position	描述车辆是否接近/离开冲突区
v_i(t)	速度	speed	连接安全间距、通行时间与停止避免
u_i(t)	加速度控制	acceleration control	优化变量, 平滑性和能耗代理

t_i^f	车辆离开控制区时间	terminal exit time	高层排序传给低层控制器的关键接口
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2.2 核心算法架构 / Core Algorithm Layout

高层监测车辆进入、延误和可行性破坏；当扰动出现时，按照优先级规则更新局部通行序列，再把目标离开时间传给每辆车的解析/数值最优控制器。低层控制器检查速度、加速度、追尾和横向冲突约束是否可执行。

The upper layer detects event-triggered infeasibility, resequences vehicles under priority rules, and sends target exit times to decentralized optimal controllers that enforce motion and safety constraints.

2.3 收敛性、稳定性或实时性 / Convergence, Stability, Real-Time Feasibility

实时性来自局部事件触发与低维车辆模型；安全性依赖硬约束而非学习策略。它不证明全局最优，但把全局重排问题约束在小窗口内，适合在线运行。

Real-time behavior comes from event-triggered local resequencing and low-dimensional dynamics; safety is constraint-based, while global optimality is intentionally relaxed.

2.4 潜在假设与边界条件 / Assumptions & Boundary Conditions

- 所有车辆为联网自动驾驶车辆 (connected automated vehicles, CAVs)，且路径已知。
- 车辆动力学可由简化双积分模型 (double-integrator model) 充分描述。
- 通信、定位和控制执行误差较小，扰动可被及时探测。

3 实验验证与结果评判 / Experimental Validation & Result Evaluation

Baselines & Environment / 基准与环境：通常与固定先来先服务/不重排序 (FCFS / no resequencing) 策略比较，重点观察扰动后恢复效率。

Validation / 验证：Numerical simulation of signal-free intersection coordination under resequencing and replanning.

Metrics / 指标：通行时间 (travel time)、延误 (delay)、停止次数 (number of stops)、控制能量代理 (control effort) 和安全约束违反。

Ablation / 消融：论文更像机制验证而非完整消融；最应补充的是关闭优先级、关闭事件触发和扩大重排窗口的对比。

Reviewer reading / 审稿式阅读提醒：阅读实验时不要只看平均值；应检查交通密度、车流方向、随机种子、失败案例和计算耗时。For doctoral reading, inspect density regimes, demand patterns, random seeds, failure cases, and computation time rather than only mean improvements.

3.1 图表阅读线索 / Figure and Table Reading Cues

PDF extraction detected approximately 11 figure references and 0 table references. Exact captions remain in the original PDF; the bilingual cues below tell you what to inspect first.

图表名称	Figure/Table Name	Reading focus / 阅读重点
系统框架图	system framework diagram	先确认输入、决策层、控制层和安全约束之间的数据流。
冲突关系图 / 通行顺序图	conflict-relation or passing-order graph	把节点、边类型和先后约束翻译成 JSSP 的作业、机器和析取约束。
实验指标对比图表	experimental metric comparison plots/tables	重点看平均值之外的密度变化、失败场景、计算时间和方差。

4 审稿人视角：批判性思考与未来方向 / Critical Thinking & Future Directions

4.1 论文局限性 / Limitations & Weaknesses

- 对全 CAV 环境依赖强，混合交通中人类驾驶车辆 (HDVs) 会破坏可行性假设。
- 优先级规则可能引入公平性问题，长期低优先级车流需要饥饿避免机制。
- 简化动力学忽略执行器延迟、轮胎约束和感知误差累积。

4.2 博士课题拓展点 / Research Extensions for PhD

- 把固定优先级扩展为学习型优先级函数 (learned priority function)，并加入公平性正则。
- 将重排序窗口映射为 JSSP machine queue，用图神经网络 (GNN) 预测候选交换收益。
- 在低层加入鲁棒 MPC 或 CBF 安全过滤器，处理通信延迟和轨迹预测误差。

4.3 如何服务当前课题 / Use in This Project

Directly supports this project’s hierarchy: a high-level sequence or schedule provides target times, and a low-level controller checks executable motion.

5 交互式可视化与 PDF 排版蓝图 / Interactive Visualization & PDF Layout Blueprint

动态参数 / Parameter: 重排序窗口大小 / Resequencing window size, range [2, 12] veh.
窗口增大通常降低延误但提高计算量，并可能增加局部公平性风险。
A larger window usually reduces delay but increases computation and fairness risk.

5.1 HTML 结构化布局建议 / HTML Structuring

- 顶部固定 metadata bar: 题名、作者、年份、主题、PDF 页数。
- 左侧目录/section navigator, 右侧为五大精读模块。
- 公式卡片使用 <pre> 或 LaTeX block, 紧跟符号表。
- 审稿人批注用 reviewer-note block, 博士拓展用 research-idea list。
- 交互参数用 input[type=range] + canvas 曲线, 打印 PDF 时隐藏交互控件但保留解释。

6 来源锚点与逐节阅读路径 / Source Anchors & Section-by-Section Reading Map

Page	Extracted heading
p.1	I. INTRODUCTION
p.1	II. M ODELING FRAMEWORK
p.1	III. Finally, we provide simulation results in Section IV, and

p.1 I. INTRODUCTION 定位问题背景、研究缺口和为什么现有保守/非协同方法不足。/ Frames the motivation, research gap, and limits of conservative or non-cooperative methods.

p.1 III. Finally, we provide simulation results in Section IV, and 检验方法是否真的改善效率、安全或能耗; 要关注基准是否公平。/ Tests efficiency, safety, or energy claims; inspect whether baselines are fair.

p.1 II. M ODELING FRAMEWORK 给出算法流程和求解机制, 应重点追踪输入、输出和安全接口。/ Gives the algorithmic mechanism; track inputs, outputs, and safety interfaces.

p.3 III. A PRIORITY-AWARE RESEQUENCING 作为局部论证段落阅读, 关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.5 IV. SIMULATION RESULTS 检验方法是否真的改善效率、安全或能耗; 要关注基准是否公平。/ Tests efficiency, safety, or energy claims; inspect whether baselines are fair.

p.6 V. CONCLUDING REMARKS AND DISCUSSION 总结贡献和未来方向, 也常暴露作者承认的边界条件。/ Summarizes contributions and often reveals author-recognized boundaries.